TOPIC 6

Overfitting and Underfitting: Explained

The original class slides, with a plain explanation under each one

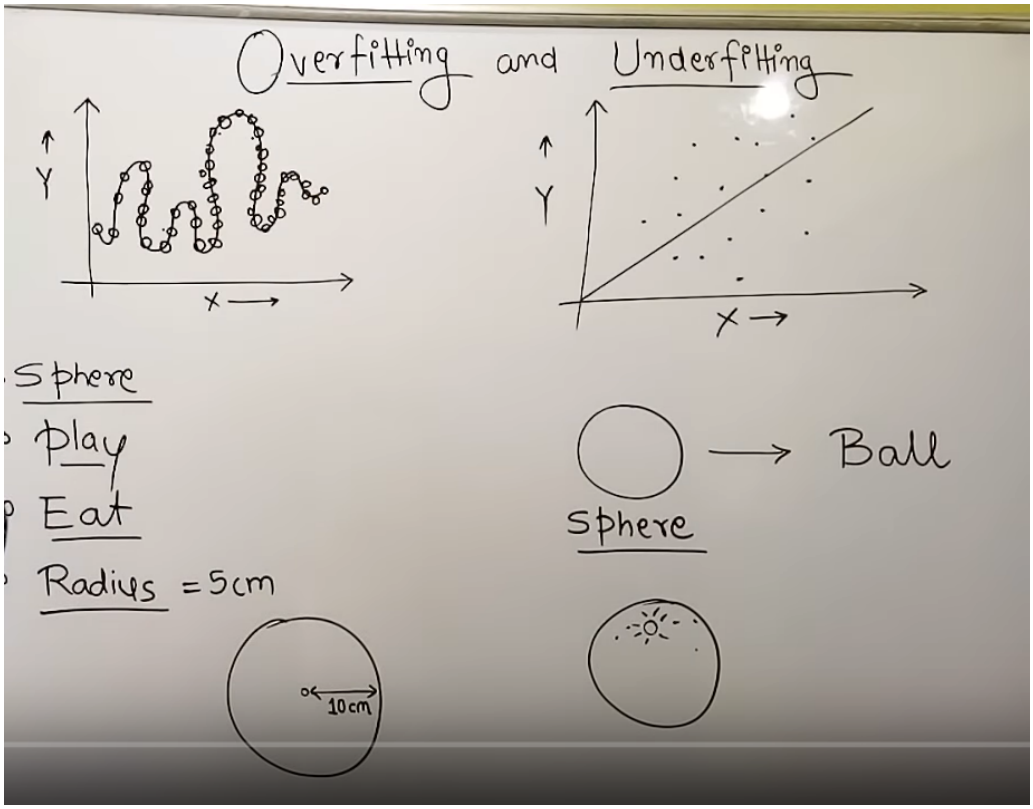
> **NOTE** **How to read this document**
>
> - Every original slide is kept exactly as it was, with its slide number under it
> - Under each slide (or small set of slides) you get a plain explanation: **what you are looking at** and **what it means**
> - The slides come from a recorded class, so a few look like screenshots of a video: that is normal

A model can fail in two opposite ways. It can be **too simple** to learn the pattern (**underfitting**) or **so flexible** that it memorises the data, including its noise (**overfitting**). These four slides show both failures, their causes, and how to fix them.

Words you will meet

Word	Plain meaning
Training data	The examples the model learns from
Validation / test data	New examples kept aside to check the model honestly
Generalise	Work well on data the model has never seen
Noise	Random variation in the data that does not follow any pattern
Overfitting	Great on training data, poor on new data (too complex)
Underfitting	Poor on both training and new data (too simple)
Regularization	Adding a penalty for complexity to keep the model simple
Cross-validation	Testing the model on several different splits of the data

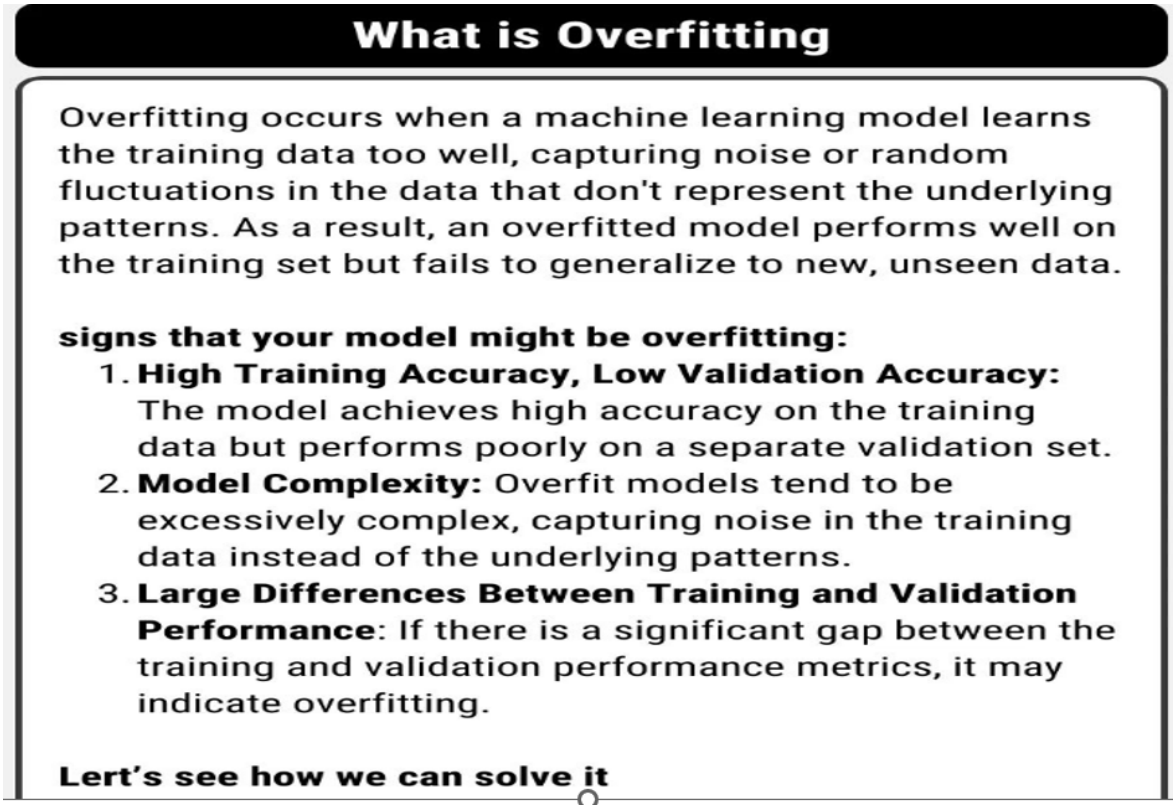
Slide 1: Whiteboard: the two failures drawn as pictures



Slide 1

Left graph (overfitting): the curve twists and turns to pass through every single dot, including the random ones. It fits the training data perfectly but it is following noise, not the real trend.
Right graph (underfitting): a straight line through scattered dots that clearly do not lie on a line. It is too simple to describe them. The sphere sketches at the bottom use a ball-versus-orange example to make the same point: a description that is too specific (play, 10 cm, one exact ball) fails to recognise other balls, while a description that is too loose (any round thing) cannot tell a ball from an orange.

Slide 2: What is overfitting? (definition and warning signs)

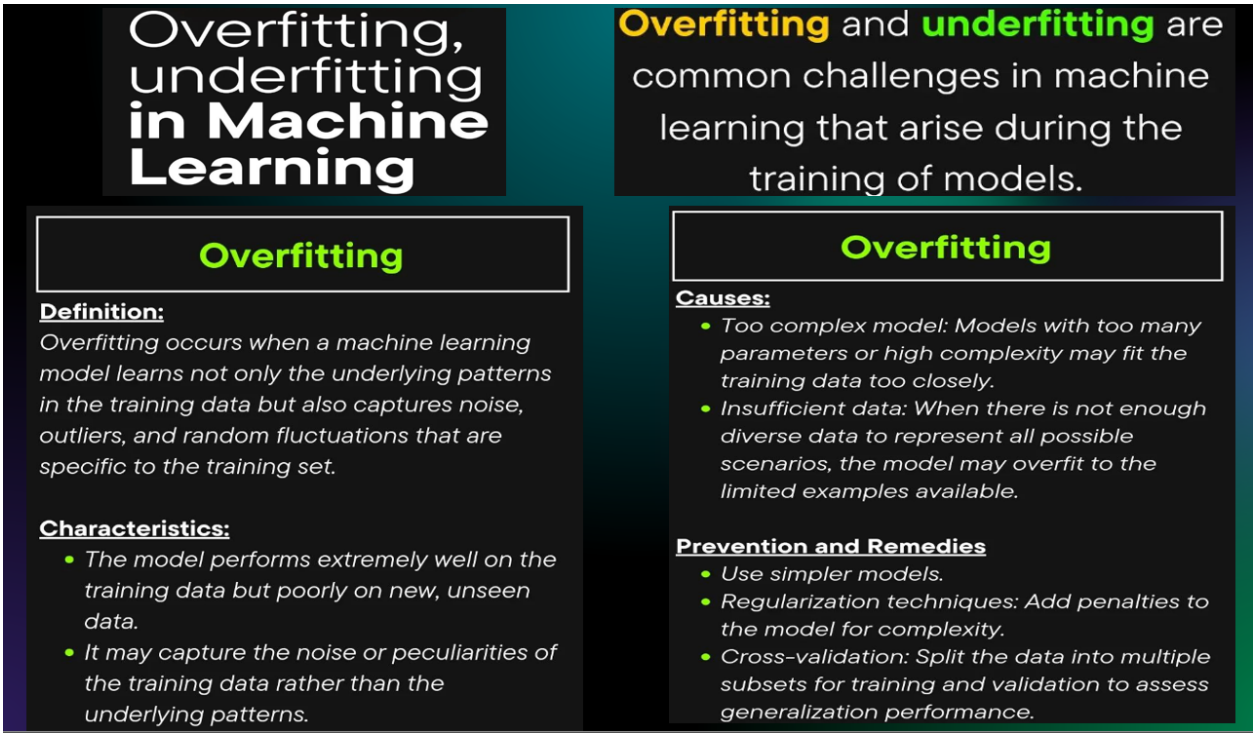


Slide 2

Overfitting happens when a model learns the training data **too well**, capturing noise and random fluctuations that do not represent the real pattern. It then does well on the training set but **fails to generalise to new, unseen data**. The slide lists three signs:

- **High training accuracy, low validation accuracy:** great on data it has seen, poor on a separate check set
- **High model complexity:** an overly complicated model is capturing noise
- **Large gap between training and validation performance:** a big difference between the two scores

Slide 3: Overfitting: definition, characteristics and causes

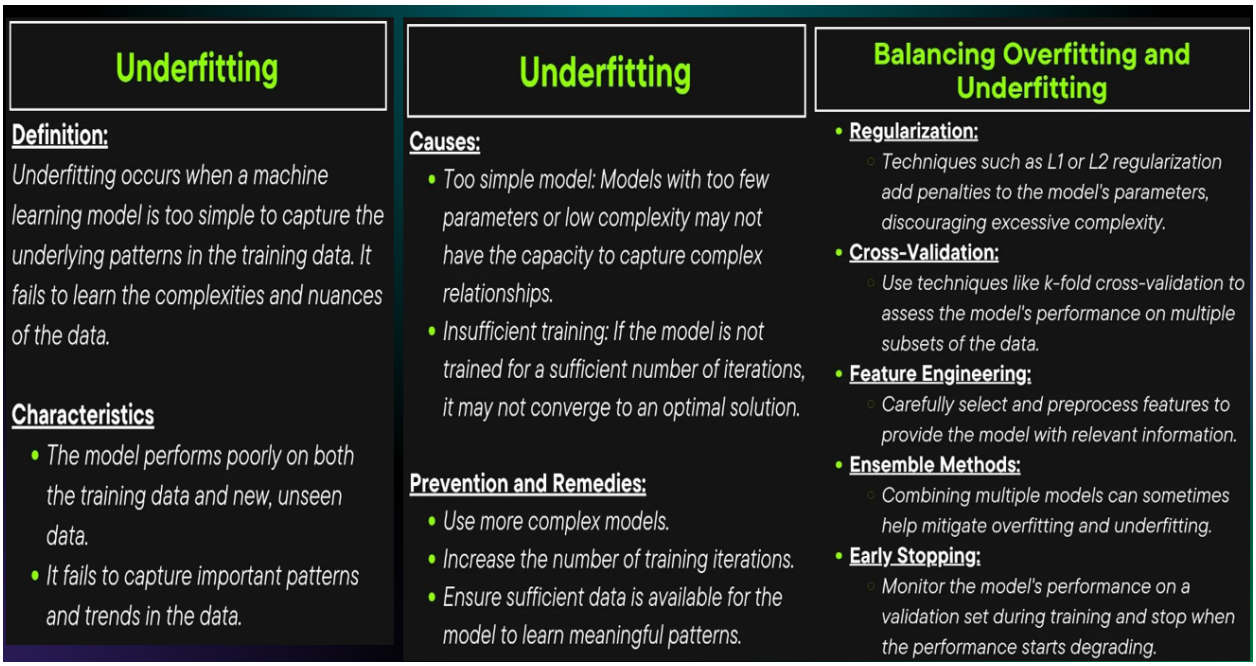


Slide 3

This slide summarises overfitting. **Definition:** the model learns not only the real patterns but also noise, outliers and random fluctuations in the training set. **Characteristics:** it performs extremely well on the training data but poorly on new data. **Causes:** a model that is too complex (too many parameters), or too little data to represent all possible situations, so the model fits the few examples it has. **Prevention:**

- Use simpler models
- Regularization: add penalties for complexity
- Cross-validation: split the data several ways to check performance on new data

Slide 4: Underfitting, and how to balance both



Slide 4

Underfitting (left and middle panels): the model is **too simple** to capture the patterns, so it performs **poorly on both the training data and new data**. Causes: a model with too few parameters, or too little training (not enough iterations). Fixes: use a more complex model, train for more iterations, make sure there is enough data. **Balancing over- and underfitting** (right panel) lists the tools:

- **Regularization:** L1 or L2 penalties discourage excessive complexity
- **Cross-validation:** such as k-fold, to assess performance on several subsets
- **Feature engineering:** carefully choose and prepare features
- **Ensemble methods:** combine several models
- **Early stopping:** stop training when the validation score starts getting worse